Calculus with Parametric and Polar Curves

ANSWER KEY

Parametric derivatives and arc length

1 For the curve $x = t^2$, $y = t^3$, find $\frac{dy}{dx}$ in terms of t.
$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3t^2}{2t} = \frac{3t}{2}$ (for $t \neq 0$).

2 Find the arc length of the curve $x = 3t^2$, $y = 2t^3$ for $0 \le t \le 2$.
$\frac{dx}{dt} = 6t$, $\frac{dy}{dt} = 6t^2$.
$L = \int_0^2 \sqrt{36t^2 + 36t^4}\, dt = \int_0^2 6t\sqrt{1 + t^2}\, dt = [2(1 + t^2)^{3/2}]_0^2 = 2(5^{3/2} - 1) = 10\sqrt{5} - 2$.

Polar area

3 Find the area enclosed by one loop of the rose $r = 3\sin 2\theta$ (for $0 \le \theta \le \pi/2$).
$A = \frac{1}{2}\int_0^{\pi/2} 9\sin^2 2\theta\, d\theta = \frac{9}{2}\int_0^{\pi/2} \frac{1 - \cos 4\theta}{2}\, d\theta = \frac{9}{4}\left[\theta - \frac{\sin 4\theta}{4}\right]_0^{\pi/2} = \frac{9\pi}{8}$.

4 Find the area enclosed by the cardioid $r = 2 + 2\cos\theta$.
$A = \frac{1}{2}\int_0^{2\pi} (2 + 2\cos\theta)^2\, d\theta = \frac{1}{2}\int_0^{2\pi} (4 + 8\cos\theta + 4\cos^2\theta)\, d\theta$. The $\cos\theta$ term integrates to 0 over a full period, and $\int_0^{2\pi} 4\cos^2\theta\, d\theta = 4\pi$: $A = \frac{1}{2}(8\pi + 4\pi) = 6\pi$.